

Agency Exercises (V2): Solutions

Agent Foundations module
Iliad Intensive

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Notation follows the exercise sheet: L is a fixed consistent formal system, $L \vdash \varphi$ means φ is provable in L , K is a fixed contradiction (so consistency is $L \not\vdash K$), and $\Box P$ is the statement, internal to L , that “ P is provable”, equivalently that $\text{ProofSeeker}(P)$ halts. The bridge: if a program halts then L proves it halts; if $L \vdash P$ then $\text{ProofSeeker}(P)$ halts.

1 Exercise 1: Gödel’s second incompleteness theorem

Recall $Z(A)$ searches for an L -proof of $\neg \text{Halts}(A(A))$ and halts if it finds one, and

$$G := \neg \text{Halts}(Z(Z)).$$

By construction $Z(Z)$ searches for an L -proof of G , so

$$Z(Z) \text{ halts} \iff L \vdash G. \quad (\star)$$

Part 1(a). G is true (assuming L consistent).

Solution. Two cases.

- $Z(Z)$ halts. By $(\star)$, $L \vdash G$, i.e. $L \vdash \neg \text{Halts}(Z(Z))$. But $Z(Z)$ actually halts, so by the bridge $L \vdash \text{Halts}(Z(Z))$. Then L proves both $\text{Halts}(Z(Z))$ and its negation, contradicting consistency.
- $Z(Z)$ runs forever. Then $\neg \text{Halts}(Z(Z))$, i.e. G , is true.

Consistency rules out the first case, so $Z(Z)$ runs forever and G is true. $\square$

Part 1(b). If $L \vdash \neg \Box K$ (i.e. L proves its own consistency) then L is inconsistent.

Solution. The argument of 1(a) used only the consistency of L , and each step is a finite manipulation of programs and proofs that L can formalize. Reading it inside L : “ $Z(Z)$ halts” yields $\Box G$ (by construction) and $\Box \text{Halts}(Z(Z))$ (the bridge), hence $\Box K$; contrapositively $L \vdash \neg \Box K \rightarrow G$. If $L \vdash \neg \Box K$, modus ponens gives

$$L \vdash G, \quad \text{i.e.} \quad L \vdash \neg \text{Halts}(Z(Z)).$$

But $L \vdash G$ means $Z(Z)$ ’s search finds a proof of G , so $Z(Z)$ halts; by the bridge $L \vdash \text{Halts}(Z(Z))$. Now L proves both $\text{Halts}(Z(Z))$ and $\neg \text{Halts}(Z(Z))$, so L is inconsistent. $\square$

Part 1(c). If $L \vdash \Box P \rightarrow P$ for every P , then L is inconsistent.

Solution. (i) Fix P with $L \vdash \neg P$. The contrapositive of $\Box P \rightarrow P$ is $\neg P \rightarrow \neg \Box P$, so from $L \vdash \Box P \rightarrow P$ we get $L \vdash \neg P \rightarrow \neg \Box P$. With $L \vdash \neg P$, modus ponens gives $L \vdash \neg \Box P$.

(ii) Take $P = K$. Since $\neg K$ is a tautology, $L \vdash \neg K$, so by (i) $L \vdash \neg \Box K$. Thus L proves its own consistency, and Part 1(b) makes L inconsistent. $\square$

2 Exercise 2: Löb's theorem

We use Necessitation (N) $L \vdash \varphi \Rightarrow L \vdash \Box \varphi$, Distribution (K) $L \vdash \Box(\varphi \rightarrow \psi) \rightarrow (\Box \varphi \rightarrow \Box \psi)$, and the Löb condition (4) $L \vdash \Box \varphi \rightarrow \Box \Box \varphi$.

Part 2(a). Necessitation. If $L \vdash \varphi$ there is a concrete proof of φ ; searching all strings, $\text{ProofSeeker}(\varphi)$ meets it and halts. “ $\text{ProofSeeker}(\varphi)$ halts” is exactly $\Box \varphi$, and a halting run is finite, so by the bridge $L \vdash \Box \varphi$. $\square$

Part 2(b). Distribution. A proof of $\varphi \rightarrow \psi$ turns a proof of φ into one of ψ (concatenate the two proofs and apply modus ponens). So if $\text{ProofSeeker}(\varphi \rightarrow \psi)$ and $\text{ProofSeeker}(\varphi)$ both halt, their outputs combine into a proof of ψ , whence $\text{ProofSeeker}(\psi)$ halts. This combining is a finite procedure L can carry out, so $L \vdash \Box(\varphi \rightarrow \psi) \rightarrow (\Box \varphi \rightarrow \Box \psi)$. $\square$

The Löb sentence λ satisfies $L \vdash \lambda \leftrightarrow (\Box \lambda \rightarrow C)$.

Part 2(c). $L \vdash \Box \lambda \rightarrow \Box C$.

Solution. From $L \vdash \lambda \rightarrow (\Box \lambda \rightarrow C)$:

$$\begin{aligned} \text{(N): } & L \vdash \Box(\lambda \rightarrow (\Box \lambda \rightarrow C)), \\ \text{(K): } & L \vdash \Box \lambda \rightarrow \Box(\Box \lambda \rightarrow C), \\ \text{(K): } & L \vdash \Box(\Box \lambda \rightarrow C) \rightarrow (\Box \Box \lambda \rightarrow \Box C), \end{aligned}$$

and chaining the last two, $L \vdash \Box \lambda \rightarrow (\Box \Box \lambda \rightarrow \Box C)$. By (4), $L \vdash \Box \lambda \rightarrow \Box \Box \lambda$. Propositionally, from $\Box \lambda$ we obtain $\Box \Box \lambda$ and then $\Box C$, so $L \vdash \Box \lambda \rightarrow \Box C$. $\square$

Part 2(d). Assuming $L \vdash \Box C \rightarrow C$, derive $L \vdash C$.

Solution. Chaining $L \vdash \Box \lambda \rightarrow \Box C$ (2(c)) with $L \vdash \Box C \rightarrow C$ gives $L \vdash \Box \lambda \rightarrow C$. The Löb equivalence gives the converse direction $L \vdash (\Box \lambda \rightarrow C) \rightarrow \lambda$, so $L \vdash \lambda$. By (N), $L \vdash \Box \lambda$; with $L \vdash \Box \lambda \rightarrow C$, modus ponens yields $L \vdash C$. $\square$

Part 2(e). FairBot. With $A := \text{“FairBot}_1(\text{FairBot}_2) = C\text{”}$ and $B := \text{“FairBot}_2(\text{FairBot}_1) = C\text{”}$, show $L \vdash A \wedge B$.

Solution. From the source code, FairBot_1 returns C iff it finds an L -proof that its opponent FairBot_2 returns C against it, i.e. a proof of B ; reading this off the code,

$$L \vdash \Box B \rightarrow A, \quad L \vdash \Box A \rightarrow B.$$

For any φ , $\Box(A \wedge B) \rightarrow \Box\varphi$ whenever $A \wedge B \rightarrow \varphi$ is a tautology: indeed (N) gives $\Box(A \wedge B \rightarrow \varphi)$ and (K) then gives $\Box(A \wedge B) \rightarrow \Box\varphi$. Applying this to $\varphi = A$ and $\varphi = B$,

$$L \vdash \Box(A \wedge B) \rightarrow \Box A, \quad L \vdash \Box(A \wedge B) \rightarrow \Box B.$$

Hence, assuming $\Box(A \wedge B)$: we get $\Box A$ and $\Box B$, then B (from $\Box A \rightarrow B$) and A (from $\Box B \rightarrow A$), so $A \wedge B$. Thus

$$L \vdash \Box(A \wedge B) \rightarrow (A \wedge B),$$

and Löb's theorem with $C := A \wedge B$ gives $L \vdash A \wedge B$. □

3 Exercise 3: The Complete Class Theorem

Higher reward is better; δ' *dominates* δ if $r(\delta') \geq r(\delta)$ coordinatewise with strict inequality in some coordinate. $r(\delta)$ is linear in the mixing weights, $R = \text{conv}\{r(a_1), \dots, r(a_k)\}$, and $\text{EU}(\delta, \pi) = \pi \cdot r(\delta)$.

Part 3(a)(i). An admissible $\delta = \sum_j \lambda_j a_j$ puts $\lambda_j = 0$ on every non-Pareto-optimal (dominated) a_j .

Solution. Suppose $\lambda_j > 0$ for a dominated a_j , and let δ' dominate it: $r(\delta') \geq r(a_j)$ with strict inequality in some coordinate i_0 . Replace the weight on a_j by δ' , i.e. play δ' with the probability λ_j formerly on a_j . This is a valid rule $\hat{\delta}$ with

$$r(\hat{\delta}) = r(\delta) + \lambda_j(r(\delta') - r(a_j)) \geq r(\delta),$$

and strict in coordinate i_0 (since $\lambda_j > 0$). So $\hat{\delta}$ dominates δ , contradicting admissibility. Hence $\lambda_j = 0$, i.e. $F \subseteq \text{conv}\{\text{Pareto-optimal pure actions}\}$. □

Part 3(a)(ii). Every Bayes-optimal rule under a prior π with $\pi_i > 0$ for all i is admissible.

Solution. If δ' dominated such a δ , then with $r(\delta')_{i_0} > r(\delta)_{i_0}$,

$$\text{EU}(\delta', \pi) - \text{EU}(\delta, \pi) = \sum_i \pi_i (r(\delta')_i - r(\delta)_i) \geq \pi_{i_0} (r(\delta')_{i_0} - r(\delta)_{i_0}) > 0,$$

since every term is ≥ 0 and the i_0 term is > 0 . This contradicts Bayes-optimality, so δ is admissible. □

Part 3(b). For a face $\mathcal{F} = \{\sum_{l=1}^p \mu_l r(a_{i_l}) : \mu_l \geq 0, \sum_l \mu_l = 1\}$ with $v_l := r(a_{i_l}) - r(a_{i_1})$ and $H = \text{span}\{v_2, \dots, v_p\}$: H is exactly the set of directions tangent to $\mathcal{F}$.

Solution. Any point of $\mathcal{F}$ equals $r(a_{i_1}) + \sum_{l \geq 2} \mu_l v_l$, so $\mathcal{F} \subseteq r(a_{i_1}) + H$. Take $x, x + h \in \mathcal{F}$, say $x = \sum_l \mu_l r(a_{i_l})$ and $x + h = \sum_l \mu'_l r(a_{i_l})$ with $\sum_l \mu_l = \sum_l \mu'_l = 1$. Then

$$h = \sum_l (\mu'_l - \mu_l) r(a_{i_l}) = \sum_{l \geq 2} (\mu'_l - \mu_l) v_l \in H,$$

using $\sum_l (\mu'_l - \mu_l) = 0$ to eliminate the $r(a_{i_1})$ term. Hence moving within $\mathcal{F}$ requires $h \in H$. Conversely, from any relative-interior point ($\mu_l > 0$) every $h \in H$ is realized: $x + \varepsilon h \in \mathcal{F}$ for small $\varepsilon > 0$. So the tangent directions of $\mathcal{F}$ are precisely H . □

Part 3(c). Let $\pi \perp H$ be normalized so $\pi_i > 0$ and $\sum_i \pi_i = 1$, and assume no point of R scores strictly higher under π than the points of $\mathcal{F}$.

Solution. (i) For $x, x' \in \mathcal{F}$ we have $x - x' \in H$ by 3(b), so $\pi \cdot (x - x') = 0$. Thus $\pi \cdot r(\delta)$ equals a common value c for all δ with $r(\delta) \in \mathcal{F}$.

(ii) Picture π as the *outward* normal of the supporting hyperplane $\{x : \pi \cdot x = c\}$: by assumption the whole risk set lies on the inner side, $\pi \cdot x \leq c$ for all $x \in R$, touching the hyperplane exactly along the face $\mathcal{F}$. Fix any $r(\delta) \in \mathcal{F}$ (so $\text{EU}(\delta, \pi) = c$) and any other rule δ' . The step from the face point $r(\delta)$ to $r(\delta') \in R$ heads back into the risk set, i.e. against the outward normal, so its dot product with π is non-positive:

$$\pi \cdot (r(\delta') - r(\delta)) \leq 0, \quad \text{equivalently} \quad \text{EU}(\delta', \pi) \leq \text{EU}(\delta, \pi).$$

Since this holds for every δ' , each rule with $r(\delta) \in \mathcal{F}$ maximizes expected utility, i.e. is Bayes-optimal under π .

Conclusion. If δ is admissible then $r(\delta) \in \mathcal{F}$ lies on some face $\mathcal{F}$; the prior π built from $\mathcal{F}$ has all $\pi_i > 0$ and, by (ii), makes δ Bayes-optimal. Conversely, by 3(a)(ii) every Bayes-optimal rule under a strictly positive prior is admissible. The two classes coincide. $\square$

4 Exercise 4: The Do-Divergence Theorem

Write $\Pr[X, A, O] = \Pr[X \mid A, O] \Pr[A \mid O] \Pr[O]$ and the blind baseline $\Pr[X, A, O \mid \text{do}(A)] = \Pr[X \mid A, O] \Pr[A] \Pr[O]$.

Claim. $D_{\text{KL}}(\Pr[X] \parallel \Pr[X \mid \text{do}(A)]) \leq \text{MI}(A; O)$.

Solution. Compute the divergence between the full joints. The factors $\Pr[X \mid A, O]$ and $\Pr[O]$ cancel in the log-ratio:

$$D_{\text{KL}}(\Pr[X, A, O] \parallel \Pr[X, A, O \mid \text{do}(A)]) = \sum_{x, a, o} \Pr[x, a, o] \log \frac{\Pr[a \mid o]}{\Pr[a]}.$$

The summand depends only on a, o ; marginalizing x and using $\Pr[a \mid o] / \Pr[a] = \Pr[a, o] / (\Pr[a] \Pr[o])$,

$$= \sum_{a, o} \Pr[a, o] \log \frac{\Pr[a, o]}{\Pr[a] \Pr[o]} = \text{MI}(A; O).$$

Marginalizing the two joints down to X sends them to $\Pr[X]$ and $\Pr[X \mid \text{do}(A)] = \sum_{a, o} \Pr[X \mid a, o] \Pr[a] \Pr[o]$ respectively, so monotonicity of KL under marginalization gives

$$D_{\text{KL}}(\Pr[X] \parallel \Pr[X \mid \text{do}(A)]) \leq D_{\text{KL}}(\Pr[X, A, O] \parallel \Pr[X, A, O \mid \text{do}(A)]) = \text{MI}(A; O).$$

5 Exercise 5: Channel Additivity

The channel factorizes as $P[Y \mid X] = P[Y_1 \mid X_1]P[Y_2 \mid X_2]$, and $P[Y, X] = P[Y \mid X]P[X]$.

Part 5(a). $\text{MI}(X; Y) = \text{MI}(X_1; Y_1) + \text{MI}(X_2; Y_2) - \text{MI}(Y_1; Y_2)$.

Solution. Using $P[x, y] = P[y | x]P[x]$ and the factorization,

$$\text{MI}(X; Y) = \sum_{x, y} P[x, y] \log \frac{P[y | x]}{P[y]} = \sum_{x, y} P[x, y] \log \frac{P[y_1 | x_1]P[y_2 | x_2]}{P[y_1, y_2]}.$$

The three terms on the right are, respectively,

$$\begin{aligned} \text{MI}(X_1; Y_1) &= \sum P[x, y] \log \frac{P[y_1 | x_1]}{P[y_1]}, \quad \text{MI}(X_2; Y_2) = \sum P[x, y] \log \frac{P[y_2 | x_2]}{P[y_2]}, \\ \text{MI}(Y_1; Y_2) &= \sum P[x, y] \log \frac{P[y_1, y_2]}{P[y_1]P[y_2]}, \end{aligned}$$

each written over the full joint (the first depends only on x_1, y_1 , etc.). Then

$$\text{MI}(X_1; Y_1) + \text{MI}(X_2; Y_2) - \text{MI}(Y_1; Y_2) = \sum_{x, y} P[x, y] \log \frac{P[y_1 | x_1]P[y_2 | x_2]}{P[y_1, y_2]} = \text{MI}(X; Y). \quad \square$$

Part 5(b). For $Q[X_1, X_2] := P[X_1]P[X_2]$, $\text{MI}_Q(X; Y) \geq \text{MI}_P(X; Y)$.

Solution. Q keeps the marginals $Q[X_i] = P[X_i]$ and the channel, so $\text{MI}_Q(X_i; Y_i) = \text{MI}_P(X_i; Y_i)$ for $i = 1, 2$ ($\text{MI}(X_i; Y_i)$ depends only on $P[X_i]$ and $P[Y_i | X_i]$). Under Q the inputs are independent, so the outputs are too:

$$Q[y_1, y_2] = \sum_{x_1, x_2} Q[x_1]Q[x_2]P[y_1 | x_1]P[y_2 | x_2] = Q[y_1]Q[y_2],$$

hence $\text{MI}_Q(Y_1; Y_2) = 0$. Applying 5(a) under each distribution,

$$\text{MI}_Q(X; Y) = \text{MI}_P(X_1; Y_1) + \text{MI}_P(X_2; Y_2) = \text{MI}_P(X; Y) + \text{MI}_P(Y_1; Y_2) \geq \text{MI}_P(X; Y),$$

the last step using $\text{MI}_P(Y_1; Y_2) \geq 0$. $\square$

Part 5(c). A throughput-maximizing input with $X_1 \perp X_2$ always exists.

Solution. The input simplex is compact and $\text{MI}(X; Y)$ continuous, so a maximizer P^* exists. Let $Q^* := P^*[X_1]P^*[X_2]$. By 5(b), $\text{MI}_{Q^*}(X; Y) \geq \text{MI}_{P^*}(X; Y) = \max$, so Q^* is also a maximizer, and under Q^* the inputs are independent. $\square$